
WHEN DO LATENT WORLD MODELS REMEMBER? A CONTROLLED AUDIT OF LEWM AND DINO-WM

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ABSTRACT

Short-context latent world models can predict local dynamics yet discard evidence needed later. We separate four questions: is an old cue required, encoded while visible, retained beyond raw context, and used in execution? We attach parameter-matched persistent modules, or *carriers*, to frozen released LeWorldModel hosts and train them on next-feature prediction without task labels. In matched four-color Reacher and PushT tasks, cue age is 4, 8, or 15. At age 15, fixed-trust leads state-space by 6.25 and 6.04 percentage points (pp); their cross-host difference is -0.21 pp (90% interval $[-4.83, 4.42]$ pp) and meets the registered ± 5 -pp equivalence criterion. Every learned arm’s linear-readout accuracy declines with age. Predeclared DINO-WM PushT and PointMaze extensions test the same intervention on spatial-token hosts with a short-context reset and an external execution test. On PushT, only state-space clears both age-15 contrasts for six-way binding; on PointMaze, every learned carrier exceeds no state and a distribution-shifting context reset, and raises fixed-controller success by up to 53.6 pp over no state. The contribution is a protocol separating retention from local prediction, decodability, and native planning—not a universally superior architecture.

1 INTRODUCTION

Latent world models compress visual observations and predict how their latent state changes under action. This makes planning efficient, but it leaves a basic question unanswered: when a decision depends on an observation that has already left the model’s input window, does the model still retain it? Low next-step prediction error does not answer that question. A target can be irrelevant to local dynamics yet decisive for a later goal, and a predictor can fit every intervening frame while discarding it.

Three common evaluations blur this distinction. Longer context exposes more raw observations but is not persistent episode state. Longer imagination extrapolates farther from its current anchor but cannot recover evidence absent from that anchor. A probe can decode a state without showing that a decision uses it. Recurrent and state-space models, retrieval, and geometric memories provide useful mechanisms (Hafner et al., 2025; Samsami et al., 2024; Deng et al., 2023; Zheng et al., 2026); the interface alone does not show which old evidence survives or affects behavior.

We introduce a controlled audit with separately predeclared, fail-closed branches that test four ordered claims. First, the old cue must be necessary: paired episodes share their late observations and actions but require different answers. Second, the frozen host must encode the cue while it is visible. Third, a small persistent state module attached to the frozen feature stream—a *carrier*—must retain the cue after it leaves raw context. Fourth, the same fixed consumer applied to every carrier arm must convert retained state into a changed executed outcome. Each stronger test runs only after its prerequisites pass. Before each branch begins, its thresholds, endpoints, seed grid, and bootstrap rule are locked; a failed branch stops instead of being resplit or relaxed.

For each admitted branch, the intervention freezes a released LeWorldModel (LeWM) (Maes et al., 2026) or DINO-WM (Zhou et al., 2025) checkpoint and trains only a parameter-matched causal carrier. Labels never enter carrier training: every learned arm optimizes the same frozen-host next-feature objective. The design compares no persistent state, GRU, LSTM, diagonal state-space, and fixed-trust predict-correct carriers. Reads occur before the decision observation, and the DINO-WM protocols additionally compare full-episode state with the same carrier reset to the legal three-frame

an executed outcome. “Full state minus no state” measures the total carrier benefit. For DINO-WM, “full state minus context reset” estimates, under the reset intervention, the contribution of carrier state accumulated before the legal three-frame predictor window while holding carrier parameters fixed.

These claims form a ladder, not interchangeable metrics. An unencoded cue cannot diagnose memory. A retained cue need not appear in the frozen predictor’s preferred coordinates. A linearly readable state need not be used by a native planner. Conversely, a small absolute read can still cause a paired change in a fixed consumer; we report both quantities so that relative sensitivity is not mistaken for competent control.

Relation to world-model memory. LeWM and DINO-WM predict from bounded latent histories (Maes et al., 2026; Zhou et al., 2025), while Fast-LeWM accelerates imagination without adding state updated by later real observations (Gao & Xu, 2026). DreamerV3, Recall to Imagine, and S4WM maintain recurrent or structured dynamics (Hafner et al., 2025; Samsami et al., 2024; Deng et al., 2023); “On Memory” studies recalled content during imagination (Laird & Clark, 2025); and Mem-World retrieves geometry-indexed visual history (Zheng et al., 2026). We do not introduce another memory family. We instead compare carriers through one frozen-host intervention and one common endpoint. RC-aux likewise shows that local predictive quality need not track planning reachability (Li et al., 2026); here the mismatch concerns evidence over real-observation time.

3 CONTROLLED AUDIT DESIGN

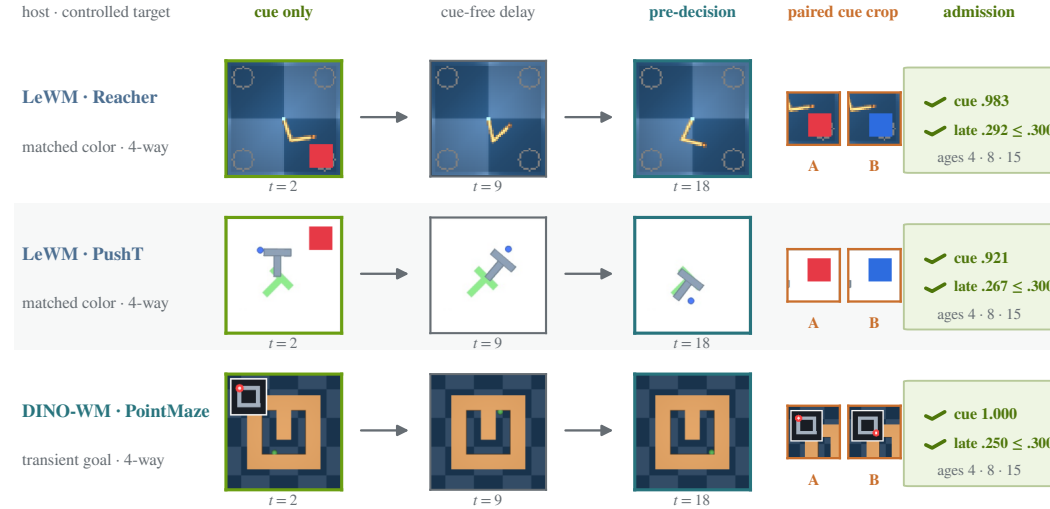


Figure 2: Task schematic for the completed matched LeWM and PointMaze studies. Green borders mark the only label-dependent pixels; the cue-free delay and teal pre-decision frames are shared across labels. Reacher and PushT use the same balanced color–location design, whereas PointMaze uses four transient mini-map goals. Admission badges summarize cue and late-input diagnostics, not retention outcomes; DINO-WM PushT cue construction is specified in Appendix D.

3.1 FROZEN HOSTS AND A COMMON CARRIER CONTRACT

The released image encoder, action encoder, latent predictor, and output projection are immutable. Each learned carrier receives the current frozen latent, the intervening action block, and its previous episode state. Its read projection is initialized at zero, so every arm begins at the released host. GRU, LSTM, diagonal state-space, and fixed-trust carriers are matched in trainable capacity within each host; optimizer, schedule, data, objective, fusion point, and state timing are shared. Fixed-trust uses a channelwise predicted state and learned input-independent correction trust. It is not privileged in the analysis: the protocol, rather than that carrier, is the methodological contribution.

LeWM consumes three 192-dimensional vector latents and a 10-dimensional action block. DINO-WM exposes 196 visual patch trajectories of width 384 together with proprioceptive and action channels. One parameter-shared carrier is applied independently to every patch trajectory; only the

frozen native predictor mixes patches. The DINO primary read is a fixed 1×1 , 2×2 , and 4×4 spatial pyramid of its final visual output. The carrier’s pre-observation prior is preserved as a secondary diagnostic.

3.2 COUNTERFACTUAL TASKS AND STOPPING GATES

Figure 2 shows the three audit settings. The completed LeWM study overlays the same four-color square on Reacher and PushT, with all 16 color–location combinations exactly balanced. The registered DINO-WM PushT protocol uses four-way token and six-way ordered-binding tasks on its native bank; their cue construction and overlay contract appear in Appendix D. The PointMaze protocol repeats each native base window under four transient mini-map goals. In the matched LeWM study the cue moves to yield ages 4, 8, and 15 at a fixed endpoint; the DINO-WM protocols place it at observations 1–3 and move the read to those same ages. Within each task, counterfactual variants share every post-cue pixel, action, simulator state, and endpoint.

Cue-time balanced accuracy must be at least .750; the matched LeWM and PointMaze protocols additionally require every class recall to reach .700. Late-input ceilings are chance plus .05: .300 for four-way tasks and .217 for six-way binding. A pixel audit requires zero changes outside the declared cue mask. Frozen host hashes are checked around every cell. Failure of any gate blocks carrier training. PointMaze also requires its simulator-controller gate before carrier training, so a failed oracle stops both retention and execution. Two earlier attempts to create a broader matched suite stopped at their declared gates and are retained rather than silently discarded. Their deterministic selection rule retained color and exact-balanced location; only then did we construct fresh banks and lock the carrier grid, endpoint, interaction, and inference before observing any new carrier outcome. The color-only comparison is therefore admission-informed rather than a fresh confirmatory test of target selection.

Table 1: Study design, not outcomes. “Included” denotes a predeclared component; “conditional” runs only after its prerequisite gates. Reset restarts the same carrier at the legal short context. Execution uses an external arm-blind consumer, never a native planner.

Study	Frozen	Cue-only	Ages	Arms	Reset	Execution
LeWM / Reacher color	Included	Included	4/8/15	5	—	—
LeWM / PushT color	Included	Included	4/8/15	5	—	—
DINO-WM / PushT token + binding	Included	Included	4/8/15	5	Included	—
DINO-WM / PointMaze goal	Included	Included	4/8/15	5	Included	Conditional

3.3 ENDPOINTS AND INFERENCE

Readouts are standardized multinomial logistic regressions trained only on the training bank and evaluated once on held-out episodes. The matched LeWM study uses 1,200/480 disjoint episodes per host, five carrier seeds, and a separate readout at each age. Its 20,000-draw paired bootstrap jointly resamples carrier seeds across hosts, independently resamples episodes within host, and stratifies by the 16 color–location combinations. The registered interaction is the PushT-minus-Reacher difference in the age-15 fixed-trust-minus-state-space contrast. Equivalence uses a two-one-sided test with a ± 5 -pp margin, hence the 90% interval; the 95% interval answers the distinct nonzero-effect question.

DINO-WM PushT uses 1,200/480 native episodes and five seeds for each task and carrier; its paired bootstrap resamples matched seeds and class-stratified held-out episodes. PointMaze uses 300/120 native base windows, expands each under all four labels, and cluster-resamples intact four-label sets by native episode as well as matched carrier seed. No-state copies are deterministic alignment artifacts and count as one independent model. The protocols preserve absolute accuracy, full-minus-no-state, full-minus-reset, validation predictions, next-feature error, and per-seed arrays; no heterogeneous score is pooled across hosts.

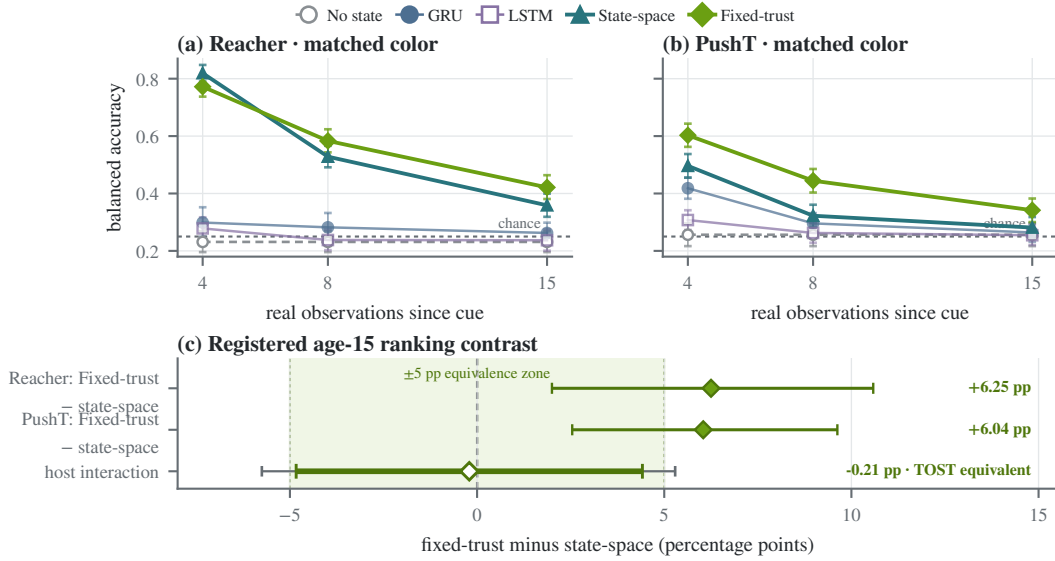


Figure 3: Matched LeWM result. Top: held-out balanced accuracy for the same four-color task and cue ages on both hosts (95% bootstrap intervals under the paired resampling design). Bottom: the registered age-15 fixed-trust-minus-state-space contrasts and their host interaction. Thin interaction whiskers are the 95% interval; thick whiskers are the 90% TOST interval used for the preregistered ± 5 -pp equivalence test.

4 MATCHED CUE AND AGE DO NOT REPRODUCE THE EARLIER RANKING REVERSAL

The motivating LeWM comparison used different cues and delays in Reacher and PushT and appeared to reverse the ranking of fixed-trust and state-space carriers. To test whether that ranking is reproduced under control, we use one four-way color target on both hosts, balance cue location as an exact nuisance, and move the cue to ages 4, 8, and 15 while holding each episode endpoint fixed. A single age-balanced carrier is trained per host, arm, and seed. All admissions are recomputed on fresh banks; metrics from the two stopped screens select the target but never enter this study’s inference.

Figure 3 gives the complete result. At age 4, state-space and fixed-trust readout accuracies are 81.9% and 77.2% on Reacher, and 49.6% and 60.3% on PushT. By age 15 they fall to 35.9%/42.1% and 28.1%/34.2%. Every learned carrier has a positive, interval-resolved paired readout decline from age 4 to 15 on both hosts: Reacher declines range from 3.63 pp (GRU) to 46.04 pp (state-space), and PushT declines from 5.29 pp (LSTM) to 26.13 pp (fixed-trust). The no-state read is invariant by construction (23.1% Reacher; 25.6% PushT).

The age-15 fixed-trust-minus-state-space contrast is +6.25 pp [2.00, 10.58] on Reacher and +6.04 pp [2.54, 9.63] on PushT. Their registered cross-host interaction is -0.21 pp. Its 90% interval $[-4.83, 4.42]$ pp lies wholly inside the ± 5 -pp equivalence region, whereas its 95% interval $[-5.75, 5.29]$ pp overlaps both equivalence boundaries and zero. Under the registered resampling model, this specific ranking contrast therefore meets the predeclared equivalence criterion; this is neither proof that the hosts are identical nor universal architecture equivalence. Per-location summaries retain the exact nuisance balance, and the appendix reports the weakest-location result for every cell.

The registered ranking interaction from the unmatched study is not reproduced after matching cue semantics and age. This does not identify the cause of the earlier reversal: the released checkpoints still differ in environment, background, dynamics, and training distribution. The supported conclusion is narrower: carrier rankings from unmatched tasks cannot be attributed to architecture alone, and a single delayed endpoint hides substantial arm-specific age sensitivity under this training contract.

5 DINO-WM PUSHT SEPARATES INTERNAL AND HOST-EXPOSED MEMORY

We next transfer the intervention from vector-latent LeWM to the original proprioceptive DINO-WM PushT checkpoint. Figure 4 compares full state with no state and a legal-context reset as cue age grows.

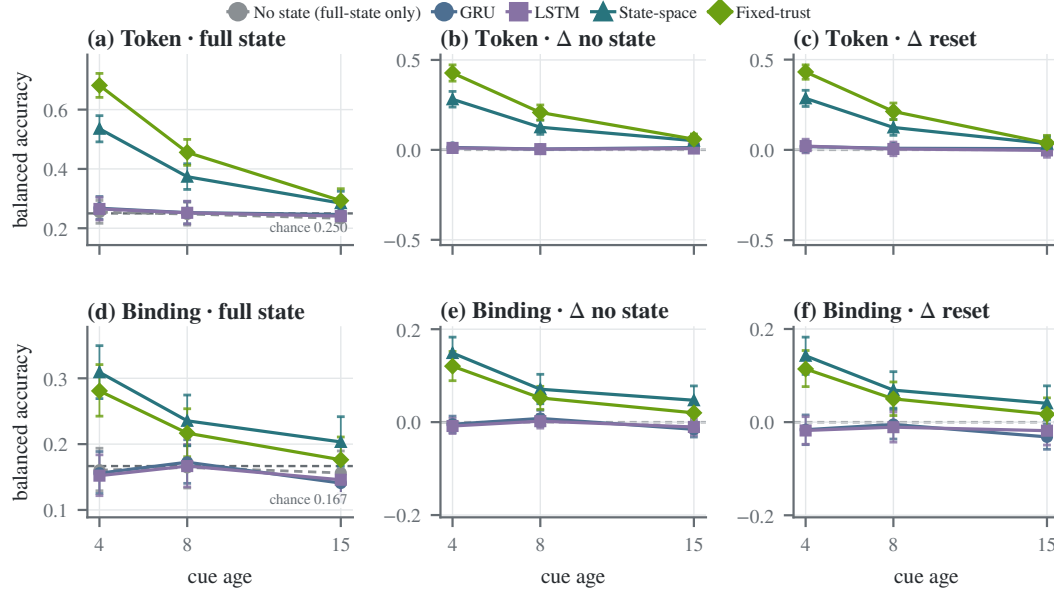


Figure 4: Persistent carriers on frozen DINO-WM PushT. Left: balanced accuracy of the frozen-predictor output; dashed lines mark chance. Middle/right: paired full-state gains over no state and legal-context reset. Whiskers are 95% matched-seed by class-stratified episode bootstrap intervals (20,000 draws). Positive intervals indicate total carrier effect or dependence on state before the reset context, respectively.

One parameter-shared carrier processes each of 196 patch trajectories; only the frozen predictor mixes patches. The four learned arms have 298,368–299,632 parameters and train without labels on 1,200 native episodes, with five seeds and 480 held-out episodes. The primary read spatially pools the frozen predictor output. Full minus no state measures total carrier effect; full minus a three-frame reset estimates dependence on state accumulated before the legal local context.

Token recall separates the carriers sharply (Figure 4). At age 4, fixed-trust and state-space reach 68.2% and 53.5%, versus 25.4% without state; their paired gains over no state are +42.7 pp [38.2, 47.2] and +28.1 pp [23.8, 32.5], and both reset contrasts resolve. At age 8, their accuracies are 45.6% and 37.4%, with positive intervals against both controls. By age 15 they fall to 29.3% and 28.5%. Gains over no state remain resolved (+6.0 pp [3.1, 8.8] and +5.1 pp [2.0, 8.3]), but neither reset interval excludes zero. GRU and LSTM remain near chance and resolve positively against neither control at any token age.

Six-way binding is harder. At age 4, state-space/fixed-trust reach 30.9%/28.1% (chance 16.7%) and beat no state by +14.9/+12.0 pp and reset by +14.3/+11.4 pp. Both comparisons remain positive at age 8. At age 15, only state-space clears both contrasts: 20.3%, +4.7 pp [1.7, 7.8] over no state, and +4.0 pp [0.5, 7.8] over reset. Fixed-trust’s no-state interval touches zero, and GRU is resolved below reset, not evidence of retention. The state-space absolute interval overlaps chance while its paired intervals are positive, illustrating the value of matched counterfactual comparisons.

The secondary carrier-prior read localizes a linear-readout exposure gap. At age 15, prior accuracy remains 99.2%/81.2% for fixed-trust/state-space on token recall and 97.2%/64.2% on binding, far above the corresponding frozen outputs. Next-feature MSE spans only .0466–.0598 and does not rank semantic recall: GRU/LSTM often minimize it while showing no gain. Thus cue identity can remain linearly accessible inside a carrier yet fail to appear in the frozen host read. This native-bank extension supports protocol and carrier portability to one DINO-WM host; it is not a matched numerical comparison with LeWM, evidence for native-planner use, or a claim that the released checkpoint itself is persistent.

6 RETAINED POINTMAZE GOALS DRIVE EXTERNAL NAVIGATION

PointMaze tests a different DINO-WM environment and MuJoCo execution. Figure 5 connects the registered retention contrasts to a shared external goal consumer and fixed waypoint controller.

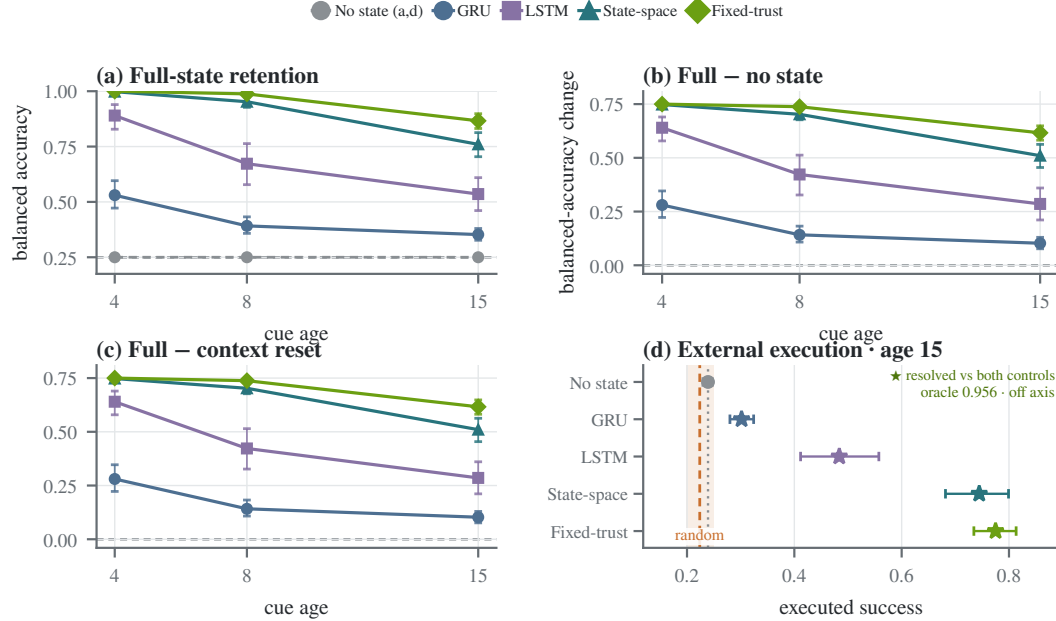


Figure 5: PointMaze retention and external use. (a) Balanced accuracy of the primary frozen-predictor read; chance and no state are 25%. (b,c) Paired full-state gains over no state and legal-context reset. (d) Age-15 executed success after a shared arm-blind consumer selects a goal for the fixed waypoint controller; stars have positive 95% lower bounds versus both no state and random goals. Intervals use 20,000 matched-seed by native-episode cluster draws; oracle success lies off axis.

We repeat 300/120 disjoint native base windows under four cue-only mini-map goals, yielding 1,200/480 sequences with identical post-cue pixels, actions, proprioception, and simulator states. Cue accuracy and minimum class recall are 100.0%; visual, action-only, and proprio-only shortcuts are 25.0%, below the .300 ceiling. The controller gate also passes: 95.6% oracle success, 93.3% for the weakest goal, zero off-diagonal false success, and exact reset replay.

All four learned carriers resolve against no state and context reset at every age (Figure 5). At ages 4/8/15, GRU reads 53.0/39.2/35.2%, LSTM 89.0/67.2/53.5%, state-space 99.8/95.2/76.0%, and fixed-trust 100.0/98.8/86.6%, while no state and the legal-context reset remain at 25.0%. Age-15 gains over no state are +10.2, +28.5, +51.0, and +61.6 pp; reset point estimates are the same. Even the smallest reset gain, GRU’s +10.2 pp [7.6, 13.0], is resolved. The complete 24 learned-arm contrasts all exclude zero, although each arm still declines with age. Reset is much more disruptive for state-space and fixed-trust: their reset/full next-visual-MSE ratios are 31.4–38.7 and 32.8–39.1 across ages (versus 1.15–1.38 for GRU and 1.85–2.40 for LSTM). Full-minus-reset therefore identifies dependence under this intervention, not an unconfounded amount of memory. Carrier-prior accuracy remains 99.6–100.0% at age 15, again separating internal retention from how much the frozen predictor exposes.

For use, one shared arm-blind multinomial consumer per seed is fitted to pooled full-state training features from every arm. At age 15 its goal accuracy is 31.7/50.9/78.4/81.5% for GRU/LSTM/state-space/fixed-trust. Executing those goals with the released waypoint controller yields 30.2/48.4/74.5/77.5% success, versus 23.9% with no state and 22.4% for realized random goals. Gains over no state are +6.2 pp [4.1, 8.5], +24.5 pp [17.2, 31.8], +50.5 pp [44.2, 56.0], and +53.6 pp [49.5, 57.4]; all also resolve over random. An independent implementation refits all five consumers and exactly reproduces their predictions. This is external memory-conditioned goal selection under one frozen checkpoint and a fixed controller, not an executed-reset comparison or native DINO-WM planning.

Table 2: Task-level claim ledger. “Resolved” lists carrier arms whose registered age-15 paired 95% interval has lower bound above zero; executed use follows its registered joint resolution flag. Entries are task-specific: no pooled score or cross-family ranking is computed.

Study	Need	Cue	No shortcut	vs. none	vs. reset	Execution
LeWM Reacher matched color	Verified	Verified	Verified	Tested; age curve	Not tested	Not tested
LeWM PushT matched color	Verified	Verified	Verified	Tested; age curve	Not tested	Not tested
DINO-WM PushT token	Verified	Verified	Verified	State-space, Fixed-trust	No arm resolved	Not tested
DINO-WM PushT binding	Verified	Verified	Verified	State-space	State-space	Not tested
DINO-WM PointMaze goal	Verified	Verified	Verified	GRU, LSTM, State-space, Fixed-trust	GRU, LSTM, State-space, Fixed-trust	GRU, LSTM, State-space, Fixed-trust

7 CROSS-STUDY ANALYSIS: MEMORY IS AN INTERFACE PROPERTY

Age changes both level and ranking. Every task shows forgetting, but not at a common rate. To compare shapes without pooling incompatible banks, consider within-task gain over that task’s own no-state arm. On DINO-WM PushT, fixed-trust token gain falls from 42.7 pp at age 4 to 6.0 pp at age 15; state-space falls from 28.1 to 5.1 pp. Their binding gains fall from 12.0 to 2.0 pp and from 14.9 to 4.7 pp. PointMaze is more durable under the same age schedule: fixed-trust retains 61.6 of its initial 75.0 pp gain, and state-space retains 51.0 of 74.8 pp. LSTM and GRU retain 28.5 of 64.0 pp and 10.2 of 28.0 pp. These are descriptive within-task ratios, not cross-bank effect estimates, but they show why one terminal delay is insufficient: an arm can lead early, converge later, or clear the reset control on one target but not another.

The carrier and frozen host form two memory stages. The DINO protocols read both the carrier’s pre-observation prior and the frozen predictor output. Their separation is large and systematic. At PushT age 15, fixed-trust retains 99.2% token and 97.2% binding accuracy internally, yet the host-exposed reads are 29.3% and 17.6%. State-space shows the same pattern at 81.2%/64.2% internally versus 28.5%/20.3% after the host. PointMaze priors remain 99.6–100.0% for every learned arm, while host-exposed accuracy spans 35.2–86.6%. Thus a low endpoint can arise because state was not retained, because the frozen fusion/predictor does not expose it, or both. The prior is not a substitute endpoint—a consumer attached there changes the interface— but it prevents these failure locations from being conflated.

Local predictive fit does not order semantic memory. On DINO-WM PushT, full-state next-feature MSE lies in a narrow .0466–.0598 range, and GRU/LSTM often attain the lowest values while their semantic output remains near chance. The PointMaze ordering is similarly misaligned at age 15: GRU has lower mean visual MSE than state-space or fixed-trust, yet its goal read is 35.2% rather than 76.0% or 86.6%. This is expected from the objective: cue identity can be irrelevant to local feature prediction after the cue disappears. It also explains why we treat next-feature loss as a shared training contract and health diagnostic, not a memory metric.

Execution identifies a consumer-specific use link. On the PointMaze deck, goal accuracy and executed success move together: 31.7/50.9/78.4/81.5% goal accuracy becomes 30.2/48.4/74.5/77.5% success. The 1.5–4.0 pp conversion loss is much smaller than the 6.2–53.6 pp gains over no state, while the oracle controller reaches 95.6%. For this fixed consumer and controller, retained goal information is therefore a major limiting factor and changing it changes behavior. The result does not imply that a different consumer, native planner, or end-to-end system would use the same coordinates. Together, the studies locate memory in the composition of target, carrier dynamics, frozen exposure interface, and consumer—not in an architecture name alone.

8 WHAT THE AUDIT CAN AND CANNOT LOCALIZE

The same intervention produces three qualitatively different diagnoses. The matched LeWM ranking interaction is not reproduced, although every learned arm loses readable information with age. On DINO-WM PushT, carrier priors remain highly decodable while frozen-predictor reads approach chance, and only state-space clears both age-15 contrasts on binding. On PointMaze, every arm

exceeds no state and a disruptive reset read, and the same external consumer converts full-state reads into simulated navigation success. These outcomes are inconsistent with a universal carrier winner or with next-feature loss as a sufficient memory score.

Full-versus-reset estimates, under that intervention, the contribution of state accumulated before the native context rather than extra parameters acting only on the final three frames. Prior-versus-output gaps localize a carrier-to-host exposure bottleneck. External execution establishes that a fixed consumer can use the representation to change outcomes. None proves that a released native planner contains or uses persistent memory, nor does the study isolate which host, task, or training factor explains the differences.

Legacy controls sharpen this boundary. On Reacher, a 56-observation raw context makes the cue readable while the trained predictor output stays near chance; in this host, local next-latent fit therefore does not guarantee semantic exposure (Appendix I). Learned rollout cohorts beat copy-last and respond to shuffled actions. That pattern is inconsistent with those two trivial baselines on these cohorts, but their anchor occurs after the old cue (Appendix I). Two label-free repairs optimize their auxiliary targets yet fail their registered support criteria (Appendix L). These are localization results, not evidence that one alternative objective would universally repair memory.

Implications for memory design and evaluation. The actionable result is a decomposition, not a carrier ranking. Local loss can prefer lower MSE and worse recall, so retention needs an incentive for old task-relevant evidence. Prior-output gaps show that the host also needs an exposure interface. Fixed-endpoint age sweeps reveal transitions hidden by one delay, while use must name its consumer: PointMaze supports an external one, not a native planner. We therefore report no-state, context-reset, internal-prior, host-output, and executed-use views instead of one pooled score.

9 LIMITATIONS

The study covers two LeWM checkpoints and two DINO-WM checkpoints, but its targets remain controlled categorical cues rather than naturally occurring long-horizon goals. The matched LeWM banks do not hold checkpoint training distribution or environment fixed, and the native DINO-WM banks are not bank-matched to LeWM, so their absolute scores must not be pooled. Parameter matching does not equal optimization matching, and freezing strengthens causal attribution while excluding end-to-end co-adaptation. Linear readouts measure accessible information, not native use. The PointMaze consumer is an external goal selector followed by a released controller; it neither tests an executed reset arm nor modifies the native DINO-WM planner. Five carrier seeds and finite held-out banks leave some rankings unresolved, so intervals and seed arrays are retained rather than reduced to winner labels. The reset itself is a distributional intervention. For state-space and fixed-trust, reset/full next-feature MSE reaches 3.85–8.25 on PushT and 31.4–39.1 on PointMaze; a full-minus-reset contrast therefore identifies dependence on pre-context state under that reset, not an unconfounded quantity of memory. The no-state and external-execution contrasts do not share this reset caveat. Ages stop at 15 real observations; multiple simultaneous items, interference, retrieval after longer delays, and end-to-end learned consumers remain outside the present claim boundary.

10 REPRODUCIBILITY AND CONCLUSION

Every formal protocol is hash-locked before outcomes. Runners reject wrong GPUs, missing cells, schema drift, altered frozen weights, and nonfinite metrics; cells commit atomically. Official verifiers check completeness and provenance, and a separate implementation reconstructs statistics from preserved predictions. The appendix reports every cell, stopped branch, amendment, bootstrap unit, and claim boundary.

Memory is an empirical chain, not a consequence of recurrence or local fit. Matching removes the earlier LeWM ranking interaction; DINO-WM shows that linear exposure can bottleneck retained state, while PointMaze converts full-state reads into external success. We therefore make bounded task-level claims, not a pooled memory score or universal architecture winner.

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A FORMAL TASK AND ENDPOINT CONTRACTS

A.1 LEGACY HETEROGENEOUS-CUE TASK INVENTORY

Table 3: Legacy heterogeneous-cue task inventory. Only the four categorical tasks enter the earlier carrier rankings in this table; the controlled matched and DINO-WM studies are reported separately below. “Observations since cue” is $q - \text{cue.off}$ at the state read before the decision observation.

Task	Checkpoint	Classes	Obs. since cue	Cue accuracy	Status
Transient-marker recall	Reacher	4	43–53	1.000	tested
Drifting-color recall	Reacher	4	43–53	1.000	tested
Transient visual-token recall	PushT	4	15	1.000	tested
Visual-binding recall	PushT	6	15	.992	tested
Occluded-target prediction	Reacher	continuous	16–20	$R^2 = .010$	stopped at availability
Shell-game capacity	Reacher	3 slots/item	43–53	.954	stopped at shortcut gate

The categorical tasks use paired counterfactuals: nuisance state, actions, physics, and every non-cue pixel are identical while the label changes. PushT additionally probes final raw latents, action blocks, and simulator states. The largest balanced accuracy is .254 for transient token recall (ceiling .300) and .171 for visual binding (ceiling .217). The decision observations are excluded by construction.

Table 4: Prerequisite checklist for the four legacy carrier-comparison tasks.

Task	Necessary	Cue encoded	No late shortcut	Pre-decision state	Cohort
Reacher marker	✓	✓ 1.000	✓ pixel-identical	✓ read 63	10 seeds + fresh
Reacher color	✓	✓ 1.000	✓ pixel-identical	✓ read 63	10 seeds + fresh
PushT token	✓	✓ 1.000	✓ .254 ≤ .300	✓ read 19	5 seeds
PushT binding	✓	✓ .992	✓ .171 ≤ .217	✓ read 19	5 seeds

A.2 TEMPORAL INTERFACE CHECKLIST

Table 5: Interfaces reported by related systems. Checks denote an explicit interface, not comparable performance. “Real-observation state” is updated as new environment observations arrive; a rollout cache alone does not qualify.

System	Bounded context	Real-observation state	Read after expiry	Frozen checkpoint swap	Multi-step rollout
LeWM (Maes et al., 2026)	✓	—	—	—	✓
Fast-LeWM (Gao & Xu, 2026)	✓	—	—	—	✓
DINO-WM (Zhou et al., 2025)	✓	—	—	—	✓
Recall to Imagine (Samsami et al., 2024)	—	✓	✓	—	✓
S4WM (Deng et al., 2023)	—	✓	✓	—	✓
On Memory (Laird & Clark, 2025)	✓	✓	✓	—	✓
Mem-World (Zheng et al., 2026)	✓	✓	✓	—	✓
This audit	✓	✓	✓	✓	diagnostic

A.3 PRE-DECISION STATE READ

For Reacher, $q = 63$ and the feature is

$$[z_{60}; z_{61}; z_{62}; b_{63}], \quad b_{63} \text{ is read before consuming } z_{63}. \quad (1)$$

For PushT, $q = 19$ and the corresponding feature is $[z_{16}; z_{17}; z_{18}; b_{19}]$. The prior may depend on earlier real latents and actions, but never on the decision observation. Temporal pooling is prohibited at the primary endpoint.

B CARRIER CONTRACT AND PARAMETER MATCHING

With frozen encoder E , frozen predictor/output map P , latent $z_t = E(o_t)$, standardized action block a_{t-1} , and carrier state m_{t-1} ,

$$(b_t, m_t, \tilde{z}_t) = C_\phi(m_{t-1}, z_t, a_{t-1}), \quad \hat{z}_{t+1} = P(\tilde{z}_{t-2:t}, a_{t-2:t}). \quad (2)$$

Only ϕ is optimized. The target is the cached raw next latent, not a carrier-fused latent. Fixed-trust memory predicts a channelwise prior from state and action, applies a learned input-independent trust to the current latent innovation, and returns a residual read to LeWM space. The GRU, LSTM, and diagonal state-space carriers use the same timing and fusion path.

Table 6: Carrier parameter and initialization contract.

Carrier	Parameters	Persistent state	Read initialization
No carrier	0	none	exact released checkpoint
GRU	75,924	learned recurrent state	zero residual
LSTM	76,372	hidden + cell state	zero residual
Diagonal state-space	76,032	diagonal recurrent state	zero residual
Fixed-trust predict-correct	76,032	mean + uncertainty tracks	zero residual

C MATCHED-COLOR PRIMARY STUDY

C.1 ADAPTIVE ORIGIN AND FRESH INFERENCE

Two broader matched-host screens were sealed before this study. Both stopped globally at their declared prerequisite gates: color was the unique target that passed at every age on both Reacher and PushT, whereas location failed at every age on both hosts and TwoRoom did not enter carrier training. We consequently locked the deterministic rule “retain color as the target and exact-balance location as a nuisance” before constructing any new bank or observing any new carrier outcome. This makes the study admission-informed rather than a fresh confirmatory test of target selection. All carrier inference, however, uses new banks and no metric from either stopped screen.

Reacher uses new simulator seeds disjoint from both screens. PushT excludes the union of every earlier HDF5 episode index before applying its new split permutation. Each host has 1,200 training and 480 validation episodes, one 20-frame sequence per episode, and disjoint train/validation episodes. Labels are independent of the base trajectory. All 16 color–location combinations are exactly balanced in each split. The four RGB colors, normalized cue centers, and square size are identical across hosts.

The cue occupies three observations. Its intervals are 12–14, 8–10, and 1–3 for ages 4, 8, and 15 at the common decision index 19. The endpoint is $[z_{16}, z_{17}, z_{18}, h_{19}^-]$; observation 19 is excluded. For each host, arm, and seed, one label-free carrier trains on an equal-age mixture. A separate training-only logistic readout is fitted per age. There are five carrier seeds per arm and 50 formal cells in total.

The paired bootstrap jointly resamples carrier seeds across hosts, resamples episodes independently within host, and stratifies episodes by the 16-way color–location label. Ages and arms remain paired within host. The primary interaction is the PushT-minus-Reacher difference in the age-15 fixed-trust-minus-state-space contrast. The equivalence margin is ± 0.05 ; the two-one-sided test therefore uses a 90% interval. We separately report a 95% interval for the nonzero-effect question. Each cell also preserves per-location and worst-location accuracy.

Table 7 reports every host–carrier–age cell, the paired age decline, and the registered interaction. Seed arrays, worst-location records, all 50 cell identities, and the independent recomputation receipt remain in the machine-readable manifest rather than an experiment-log dump.

Table 7: Matched-color retention across hosts and evidence ages.

Host	Carrier	Age 4	Age 8	Age 15	Δ_{4-15}
Reacher	No state	23.1 [19.6, 26.9]	23.1 [19.6, 26.9]	23.1 [19.6, 26.9]	+0.0 [+0.0, +0.0]
	GRU	29.9 [25.0, 35.2]	28.2 [23.7, 33.2]	26.2 [22.8, 29.8]	+3.6 [+0.6, +7.0]
	LSTM	27.8 [24.1, 31.7]	23.8 [20.3, 27.5]	23.7 [20.1, 27.5]	+4.1 [+2.0, +6.3]
	State-space	81.9 [79.0, 84.8]	52.8 [49.1, 56.6]	35.9 [31.9, 40.0]	+46.0 [+41.6, +50.5]
	Fixed-trust	77.2 [73.7, 80.7]	58.3 [54.3, 62.4]	42.1 [38.1, 46.4]	+35.1 [+30.5, +39.7]
PushT	No state	25.6 [21.7, 29.6]	25.6 [21.7, 29.6]	25.6 [21.7, 29.6]	+0.0 [+0.0, +0.0]
	GRU	41.8 [38.2, 45.6]	29.6 [26.1, 33.2]	26.4 [23.2, 29.7]	+15.4 [+11.8, +19.2]
	LSTM	30.8 [27.3, 34.2]	26.2 [22.8, 29.6]	25.5 [22.0, 28.9]	+5.3 [+2.2, +8.5]
	State-space	49.6 [45.5, 53.7]	32.2 [28.4, 36.1]	28.1 [24.5, 31.8]	+21.5 [+17.6, +25.5]
	Fixed-trust	60.3 [56.2, 64.3]	44.5 [40.3, 48.5]	34.2 [30.0, 38.2]	+26.1 [+21.9, +30.4]
Registered host ranking interaction		−0.2 [−5.7, +5.3]; 90% CI [−4.8, +4.4]; equivalent within ± 5 points			

Entries are percentage-point balanced accuracy with 95% intervals; Δ_{4-15} is the paired shortest-minus-longest contrast. The registered interaction is the PushT-minus-Reacher difference in fixed-trust-minus-state-space accuracy at age 15.

D PERSISTENT SPATIAL CARRIERS ON DINO-WM PUSH T

Unlike the diagnostic-only imagination precursor in Appendix J, this study trains and compares persistent carrier sidecars on the frozen host. The host is the original official proprioceptive DINO-WM PushT checkpoint and its native OSF `pusht_noise` bank. This bank is not matched to LeWM and must not support a numerical cross-family ranking. The frozen modules are the DINOv2 encoder, predictor, action encoder, and proprioceptive encoder. Their combined state digest is checked before and after every cell.

Each observation yields 196 visual tokens of width 384. A single parameter-shared carrier operates independently along all 196 patch trajectories; the official frozen predictor remains the only cross-patch mixing operation. The residual carrier output replaces only native visual channels. Native proprioceptive and action channels are unchanged. GRU, LSTM, state-space, and fixed-trust arms contain 298,368, 299,632, 299,520, and 299,520 trainable parameters, respectively; the no-state arm contains zero. The objective is the frozen native shifted next-feature MSE on visual and proprioceptive targets, with no task label.

Token recall has four labels: a red, green, blue, or amber cue colors both the frame border and a square in its upper-left corner. Binding recall has six labels: the six permutations of three ordered red/green/blue swatches, followed by one label-invariant neutral anchor. Only these declared cue pixels differ across counterfactuals; the underlying native frame and every post-cue input remain identical.

Token and binding cues occupy observations 1–3. Reads occur at endpoint frames 7, 11, and 18, giving ages 4, 8, and 15 while excluding the endpoint observation. The primary representation is the frozen predictor’s final visual output under full episode state, pooled at spatial levels 1, 2, 4. The reset control reinitializes the identical trained carrier at the first frame of each legal three-frame context. A gain over no state measures the total carrier effect; a gain over reset estimates, under this intervention, the contribution of state accumulated before that local context. Strong persistent retention requires both paired 95% intervals to lie above zero.

The formal grid contains two tasks, five arms, five seeds, and three ages (50 formal cells; ages share a cell). It uses 1,200 training and 480 validation episodes. Inference is a 20,000-draw matched-seed and class-stratified held-out-episode bootstrap. A sealed numerical amendment changed only the cache-layout tolerance after the parent attempt stopped before carrier outcomes: exact original-layout replay remained bit-identical, and the reusable full-bank cache drift was bounded below 10^{-4} with zero existing-readout prediction flips. Tasks, grid, objective, optimization, endpoints, and inference were unchanged; the parent stop receipt is preserved.

Table 8 gives all 30 task–age–arm reads and both paired controls. At age 15 the mean carrier-prior reads are 66.0/48.0/81.2/ 99.2% for GRU/LSTM/state-space/fixed-trust on token recall and 38.3/33.5/64.2/97.2% on binding. Full-state next-feature MSE remains within .0466–.0598 and does not rank semantic recall. The manifest binds these diagnostics, the amendment, all 50 cells, their preserved predictions, and the independent 20,000-draw recomputation.

Table 8: DINO-WM PushT carrier results on two registered visual-memory tasks.

Task	Age	Carrier	BAcc	Δ no state	Δ reset	Params
Token recall	4	No state	25.4 [21.7, 29.4]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	26.8 [23.0, 30.7]	+1.3 [−0.5, +3.2]	+1.8 [−1.3, +5.0]	298,368
		LSTM	26.5 [22.7, 30.4]	+1.1 [−0.5, +2.7]	+2.1 [−1.7, +5.9]	299,632
		State-space	53.5 [49.1, 58.0]	+28.1 [+23.8, +32.5]	+28.5 [+24.0, +33.0]	299,520
		Fixed-trust	68.2 [64.1, 72.2]	+42.7 [+38.2, +47.2]	+43.2 [+39.1, +47.1]	299,520
	8	No state	24.8 [21.0, 28.7]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	25.3 [21.5, 29.0]	+0.5 [−1.4, +2.3]	+0.9 [−1.5, +3.3]	298,368
		LSTM	25.2 [21.5, 29.1]	+0.4 [−1.2, +2.1]	+0.5 [−3.3, +4.3]	299,632
		State-space	37.4 [33.1, 41.8]	+12.6 [+8.6, +16.5]	+12.4 [+8.0, +16.8]	299,520
		Fixed-trust	45.6 [41.2, 50.0]	+20.8 [+16.6, +25.0]	+21.3 [+16.8, +26.0]	299,520
	15	No state	23.3 [19.6, 27.1]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	24.6 [20.9, 28.4]	+1.3 [−0.5, +3.1]	+0.7 [−2.7, +4.0]	298,368
		LSTM	24.1 [20.5, 27.8]	+0.8 [−0.9, +2.5]	−0.3 [−4.1, +3.6]	299,632
		State-space	28.5 [24.7, 32.5]	+5.1 [+2.0, +8.3]	+3.5 [−0.4, +7.3]	299,520
		Fixed-trust	29.3 [25.3, 33.3]	+6.0 [+3.1, +8.8]	+3.8 [−0.4, +8.0]	299,520
Binding recall	4	No state	16.0 [12.9, 19.4]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	15.6 [12.5, 18.9]	−0.5 [−2.2, +1.3]	−1.6 [−4.8, +1.5]	298,368
		LSTM	15.2 [12.2, 18.4]	−0.8 [−2.5, +0.8]	−1.8 [−4.7, +1.2]	299,632
		State-space	30.9 [26.9, 35.0]	+14.9 [+11.6, +18.3]	+14.3 [+10.3, +18.2]	299,520
		Fixed-trust	28.1 [24.2, 32.1]	+12.0 [+8.9, +15.2]	+11.4 [+7.6, +15.4]	299,520
	8	No state	16.5 [13.3, 19.8]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	17.2 [14.0, 20.7]	+0.8 [−0.9, +2.5]	−0.5 [−3.6, +2.7]	298,368
		LSTM	16.7 [13.5, 20.0]	+0.2 [−1.3, +1.7]	−1.1 [−4.3, +2.2]	299,632
		State-space	23.5 [19.8, 27.5]	+7.1 [+4.0, +10.3]	+6.9 [+3.0, +10.8]	299,520
		Fixed-trust	21.7 [18.1, 25.4]	+5.2 [+2.8, +7.7]	+5.0 [+1.4, +8.6]	299,520
	15	No state	15.6 [12.5, 19.0]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
		GRU	14.1 [11.2, 17.1]	−1.5 [−3.2, +0.1]	−3.2 [−5.8, −0.4]	298,368
		LSTM	14.6 [11.7, 17.7]	−1.0 [−2.7, +0.6]	−1.8 [−4.9, +1.3]	299,632
		State-space	20.3 [16.6, 24.2]	+4.7 [+1.7, +7.8]	+4.0 [+0.5, +7.8]	299,520
		Fixed-trust	17.6 [14.3, 21.1]	+2.0 [+0.0, +4.0]	+1.7 [−1.6, +5.2]	299,520

All performance entries are percentage points with 95% intervals. Absolute BAcc is class balanced; contrasts preserve matched carrier seeds and class-stratified held-out episodes. Params counts trainable carrier parameters only.

E DINO-WM POINTMAZE RETENTION AND EXTERNAL USE

The host is the official proprioceptive DINO-WM PointMaze checkpoint and its 2,000-episode native trajectory archive. We choose 300 training and 120 validation base windows from disjoint native episodes. Each base window is repeated under all four labels, yielding 1,200/480 sequences while preserving the native episode as the statistical cluster. The target is one of four fixed waypoints. A transient mini-map card contains a label-invariant U-shaped route and one position-varying goal dot; no label-specific color is permitted. Cue pixels occur only in observations 1–3, and all later pixels, actions, proprioception, and simulator states are identical across the four variants.

The carrier interface, ages, spatial pyramid, objective, arms, and five-seed training contract match the DINO-WM PushT extension. The 20,000-draw paired bootstrap resamples matched carrier seeds and native validation episodes while keeping each four-label counterfactual set intact. We retain absolute full-state accuracy, full-minus-no-state, full-minus-reset, prior reads, and next-feature error at ages 4, 8, and 15.

The reset is deliberately diagnostic rather than distribution preserving. Across ages, mean reset/full next-visual-MSE ratios are 1.15–1.38 for GRU, 1.85–2.40 for LSTM, 31.4–38.7 for state-space, and 32.8–39.1 for fixed-trust. A positive full-minus-reset interval therefore shows dependence on state preceding the legal context under this intervention; it is not an unconfounded scalar estimate of retained information. Full-minus-no-state and external execution are reported separately.

External execution is enabled only after cue, shortcut, frozen-host, and controller gates pass. At age 15, one standardized multinomial consumer per carrier seed is trained on the pooled full-state training features from every arm. The deck contains no arm identifier, and the same fitted consumer is evaluated on the five full-state arm sources. Prior and reset reads are not executed. Its selected waypoint is executed for 160 steps by the unmodified released PointMaze WaypointController (source SHA-256 d7b2f9a8a101) under MuJoCo 3.10.0 and the released XML (SHA-256 64d83955572d). The oracle must achieve at least .90 overall and .85 for every goal, off-diagonal false success must

not exceed .10, and deterministic reset replay must equal 1.0. A resolved execution gain requires positive paired lower bounds against both no state and the realized random-goal arm. This is external memory-conditioned navigation execution, not native DINO-WM planning.

Tables 9–11 report every gate, all 15 carrier reads with both controls, and every external consumer outcome. The machine-readable manifest binds the native episode clusters, carrier seed arrays, five independently refitted consumer coefficient digests, execution deck, all 25 cells, and both audit receipts.

Table 9: DINO-WM PointMaze persistent-carrier results.

Age	Carrier	BAcc	Δ no state	Δ reset	Params
4	No state	25.0 [25.0, 25.0]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
	GRU	53.0 [47.2, 59.6]	+28.0 [+22.2, +34.6]	+28.0 [+22.2, +34.7]	298,368
	LSTM	89.0 [82.8, 94.0]	+64.0 [+57.9, +69.0]	+64.0 [+57.9, +69.0]	299,632
	State-space	99.8 [99.3, 100.0]	+74.8 [+74.4, +75.0]	+74.8 [+74.4, +75.0]	299,520
	Fixed-trust	100.0 [100.0, 100.0]	+75.0 [+75.0, +75.0]	+75.0 [+75.0, +75.0]	299,520
8	No state	25.0 [25.0, 25.0]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
	GRU	39.2 [35.8, 43.2]	+14.2 [+10.8, +18.2]	+14.2 [+10.8, +18.2]	298,368
	LSTM	67.2 [57.8, 76.3]	+42.2 [+32.7, +51.2]	+42.2 [+32.7, +51.5]	299,632
	State-space	95.2 [92.7, 97.3]	+70.2 [+67.7, +72.3]	+70.2 [+67.7, +72.3]	299,520
	Fixed-trust	98.8 [97.8, 99.6]	+73.8 [+72.7, +74.6]	+73.8 [+72.7, +74.6]	299,520
15	No state	25.0 [25.0, 25.0]	<i>reference</i>	+0.0 [+0.0, +0.0]	0
	GRU	35.2 [32.7, 38.0]	+10.2 [+7.7, +13.0]	+10.2 [+7.6, +13.0]	298,368
	LSTM	53.5 [46.1, 61.0]	+28.5 [+21.1, +36.0]	+28.5 [+21.1, +36.0]	299,632
	State-space	76.0 [70.4, 81.3]	+51.0 [+45.5, +56.2]	+51.0 [+45.4, +56.3]	299,520
	Fixed-trust	86.6 [83.2, 89.8]	+61.6 [+58.2, +64.9]	+61.6 [+58.2, +64.8]	299,520

Entries are percentage points with 95% intervals from matched carrier-seed by equal-native-episode cluster resampling; all four counterfactual labels remain within their native episode cluster.

Table 10: PointMaze prerequisite admission and controller checklist.

Gate	Observed	Criterion	Status
Old-observation requirement	four labels; no post-cue leak	cue-only counterfactual	PASS
Visible-cue encoding	100.0 BAcc; 100.0 min recall	≥ 75.0 ; ≥ 70.0	PASS
No-cue visual shortcut	max 25.0	≤ 30.0	PASS
Action-only shortcut	max 25.0	≤ 30.0	PASS
Proprio-only shortcut	max 25.0	≤ 30.0	PASS
Cue-only invariance	outside=0; post-cue=0	both 0	PASS
Frozen host	digest unchanged	exact equality	PASS
Controller oracle	95.6	≥ 90.0	PASS
Worst-goal oracle	93.3	≥ 85.0	PASS
Off-diagonal false success	0.0	≤ 10.0	PASS
Deterministic replay	100.0	≥ 100.0	PASS

Every registered prerequisite must pass before carrier training or external-use evaluation. Shortcut values are the maximum across evidence ages; controller values are percentages.

Table 11: PointMaze external-use results under the fixed waypoint controller.

Carrier	Goal accuracy	Executed success	Δ no state	Δ random	Resolved
No state	25.0 [25.0, 25.0]	23.9 [23.0, 24.7]	+0.0 [+0.0, +0.0]	+1.5 [−1.0, +4.0]	NO
GRU	31.7 [29.3, 34.3]	30.2 [27.7, 32.7]	+6.2 [+4.1, +8.5]	+7.8 [+4.0, +11.5]	YES
LSTM	50.9 [43.2, 58.6]	48.4 [41.0, 56.0]	+24.5 [+17.2, +31.8]	+26.0 [+18.4, +33.8]	YES
State-space	78.4 [72.0, 83.7]	74.5 [68.1, 80.2]	+50.5 [+44.2, +56.0]	+52.1 [+45.9, +57.6]	YES
Fixed-trust	81.5 [78.0, 84.9]	77.5 [73.0, 81.7]	+53.6 [+49.5, +57.4]	+55.1 [+50.3, +59.8]	YES

Realized random-goal success: 22.4 [19.8, 24.9]; oracle executed success: 95.6.

Entries are percentage points with 95% native-episode cluster intervals. Consumers are shared and arm blind; execution uses the fixed released waypoint controller rather than native model planning.

F COMPLETE HETEROGENEOUS-CUE REACHER RESULTS

F.1 REGISTERED-BANK SEED EXTENSION

Table 12: Registered-bank ordinary accuracy at the final read and for the exploratory trajectory feature. GRU, state-space, and fixed-trust combine ten seeds; LSTM is the original five-seed result because it was not part of the locked extension.

Carrier	Marker final	Color final	Marker trajectory	Color trajectory
No carrier	.258	.212	.258	.212
GRU	.275 [.267,.284]	.217 [.206,.228]	.270	.234
LSTM (5 seeds)	.272 [.258,.285]	.226 [.211,.237]	.255	.226
State-space	.269 [.260,.278]	.233 [.227,.240]	.512	.420
Fixed-trust	.275 [.270,.280]	.201 [.199,.203]	.455	.414

The endpoint reversal is not sampling noise from one seed. Fixed-trust is positive relative to no state on marker recall and negative on color recall; state-space is positive on both. The trajectory feature changes the ranking because it pools all post-cue reads and is therefore exploratory.

F.2 FRESH VALIDATION BANKS

Table 13: Equal-task-bank fresh-validation ordinary accuracy, with no retraining. Means and 95% intervals pool both new banks inside each of five matched checkpoints.

Carrier	Pre-decision state read	Trajectory-feature diagnostic
No carrier	.252 [.252,.252]	.252 [.252,.252]
GRU	.259 [.252,.268]	.263 [.256,.270]
LSTM	.258 [.252,.264]	.264 [.260,.268]
State-space	.261 [.253,.269]	.448 [.432,.462]
Fixed-trust	.252 [.251,.253]	.433 [.427,.439]

Figure 6 visualizes the same fresh-bank endpoint comparison; the complete numeric values are retained above.

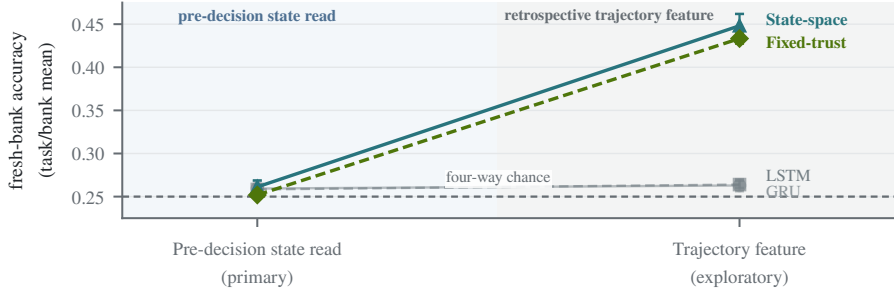


Figure 6: Fresh-bank endpoint sensitivity. Retrospective pooling over post-cue state reads is more decodable for state-space and fixed-trust, yet the pre-decision state read remains near chance. Tasks and banks are averaged within five checkpoints; whiskers are 95% seed intervals.

G COMPLETE HETEROGENEOUS-CUE PUSHT RESULTS

Table 14: Official PushT frozen-checkpoint results. Values are balanced accuracy [95% seed interval], followed by the paired change from no state.

Carrier	Token recall (chance .25)	Binding recall (chance .167)	Equal-task interpretation
No carrier	.254	.162	reference
GRU	.239 [.234,.245]; -.015	.167 [.156,.178]; +.005	negative / unresolved
LSTM	.249 [.237,.261]; -.005	.161 [.152,.173]; -.001	unresolved
State-space	.314 [.291,.338]; +.060	.175 [.167,.180]; +.013	positive on both
Fixed-trust	.650 [.646,.654]; +.396	.214 [.210,.218]; +.052	positive on both

Table 15: PushT seed-level balanced accuracy. The no-state copies are identical by construction and are not independent models.

Carrier	Token recall seeds 0–4	Binding recall seeds 0–4
No carrier	.254, .254, .254, .254, .254	.162, .162, .162, .162, .162
GRU	.242, .231, .240, .233, .250	.160, .181, .183, .152, .158
LSTM	.246, .248, .250, .229, .271	.167, .154, .150, .152, .183
State-space	.333, .308, .279, .356, .294	.177, .181, .158, .177, .181
Fixed-trust	.654, .646, .656, .648, .646	.206, .213, .215, .221, .217

The fixed-trust result is not credited as a general architecture claim because the same carrier is unfavorable on Reacher color recall. Its value is as a positive control showing that the frozen intervention can learn delayed information under the next-latent objective for some hosts and cues.

H HETEROGENEOUS-CUE FIXED-ENDPOINT PRECURSOR

This precursor retains different cue semantics across tasks; the matched-color primary comparison is reported in Appendix C. The cross-checkpoint comparison changes task, visuals, dynamics, and delay together. The strict branch intervenes on cue placement within each task by holding the decision index and underlying episode trajectory fixed while translating only the cue overlay. Reacher always reads before observation 63 at 15, 24, 32, or 43 observations after the cue; PushT always reads before observation 19 at 4, 8, or 15 observations after the cue. The final raw-latent window, actions, simulator state, label, and base episode are identical across ages. Absolute cue time and cue background co-vary with the offset. Reacher regeneration must exactly match the archived source arrays before cue translation; PushT retains the same official episode and local-start identity and permits mutations only inside declared cue pixels.

For each task, module, and seed, one memory module is trained on a mixture containing every training episode once at every registered age, with equal age weight. The same held-out base episode is then evaluated at every age. This avoids fitting a different module to each delay; a separate prespecified linear readout is nevertheless fitted on the training split at each age. All learned modules use five seeds, the frozen encoder/predictor and 100-epoch training schedule remain fixed, and the readout excludes the decision observation. Paired 95% intervals use 20,000 bootstrap draws over checkpoint seed and episode, with class-stratified episode resampling for PushT. Task admission and final evaluation use the same locked validation bank, so inference is conditional on that bank; admission is evaluated before memory training and does not select a module or age.

Table 16: Strict cue-offset admission checks. Values are the minimum cue readout and maximum endpoint shortcut over all registered ages. Every age passes before memory-module metrics are aggregated.

Task	Obs. since cue	Min. cue readout	Max. late shortcut	Status
Reacher marker	15, 24, 32, 43	1.000	.271 ≤ .300	✓
Reacher color	15, 24, 32, 43	1.000	.263 ≤ .300	✓
PushT token	4, 8, 15	1.000	.254 ≤ .300	✓
PushT binding	4, 8, 15	.985	.171 ≤ .217	✓

Pre-training cache-schema receipt. The locked PushT preparer retained a redundant `local_start` identity array after verifying exact episode/local-start pairing, whereas the locked trainer rejects arrays it does not consume. Seventeen queued commands therefore stopped at cache validation in 5–6 seconds, before creating a carrier, optimizer, checkpoint, or metric. Before any PushT carrier metric existed, a disclosed post-lock artifact-schema amendment archived all 12 original caches, removed only `local_start` from trainer-facing copies, and updated their artifact records. Labels, latents, actions, cue boundaries, and episode indices are byte-identical before and after; admissions and both locked programs are unchanged. The receipt (SHA-256 prefix `07c6581c`) hashes every failed log and original/amended artifact. This is implementation provenance, not an experimental stop or task failure.

H.1 COMPLETE FIXED-ENDPOINT RESULTS

Tables below report the full registered inferential grid. No-state entries are absolute held-out accuracy; because the late inputs and endpoint are identical, they are exactly constant across ages. Learned-

module entries are paired accuracy changes from that no-state feature on the same episodes. Reacher reports ordinary accuracy and PushT balanced accuracy. The shared carrier weights are age-balanced, but each age uses a separately fitted training-only linear readout.

Table 17: Reacher fixed-endpoint grid: no-state absolute accuracy and learned-module change from no state [paired 95% CI].

Task	Module	Age 15	Age 24	Age 32	Age 43
Marker	No state	.258	.258	.258	.258
	GRU	+0.023 [−.006,.054]	+0.024 [−.005,.055]	+0.024 [−.005,.055]	+0.024 [−.005,.055]
	LSTM	+0.018 [−.014,.050]	+0.018 [−.013,.051]	+0.018 [−.014,.050]	+0.018 [−.013,.051]
	State-space	+0.169 [.109,.228]	+0.068 [.018,.116]	+0.031 [−.013,.075]	+0.027 [−.016,.069]
	Fixed-trust	+0.136 [.073,.199]	+0.009 [−.035,.053]	+0.008 [−.037,.053]	+0.010 [−.035,.056]
Color	No state	.208	.208	.208	.208
	GRU	+0.017 [−.013,.049]	+0.018 [−.012,.049]	+0.017 [−.013,.049]	+0.017 [−.014,.050]
	LSTM	+0.007 [−.024,.038]	+0.008 [−.023,.038]	+0.009 [−.022,.040]	+0.009 [−.022,.040]
	State-space	+0.141 [.095,.189]	+0.048 [.012,.086]	+0.028 [−.005,.063]	+0.020 [−.013,.054]
	Fixed-trust	+0.243 [.175,.310]	+0.034 [−.016,.084]	−.002 [−.048,.044]	−.002 [−.048,.042]

Table 18: PushT fixed-endpoint grid: no-state absolute balanced accuracy and learned-module change from no state [paired 95% CI].

Task	Module	Age 4	Age 8	Age 15
Token	No state	.252	.252	.252
	GRU	+0.403 [.347,.462]	+0.179 [.127,.235]	+0.036 [.005,.068]
	LSTM	+0.168 [.133,.203]	+0.018 [−.011,.047]	+0.002 [−.026,.029]
	State-space	+0.667 [.625,.708]	+0.311 [.261,.361]	+0.078 [.047,.109]
	Fixed-trust	+0.694 [.652,.735]	+0.528 [.478,.578]	+0.349 [.302,.396]
Binding	No state	.163	.163	.163
	GRU	+0.113 [.076,.153]	+0.033 [.003,.064]	+0.003 [−.023,.028]
	LSTM	+0.020 [−.010,.052]	−.009 [−.035,.017]	−.004 [−.035,.027]
	State-space	+0.205 [.162,.249]	+0.043 [.014,.072]	+0.011 [−.018,.039]
	Fixed-trust	+0.225 [.180,.270]	+0.082 [.042,.121]	+0.054 [.018,.091]

Table 19: Direct shortest-minus-longest paired contrast [95% CI]. Positive values mean the earlier cue is more readable at the same endpoint.

Module	Reacher marker	Reacher color	PushT token	PushT binding
No state	.000 [.000,.000]	.000 [.000,.000]	.000 [.000,.000]	.000 [.000,.000]
GRU	−.001 [−.005,.000]	.000 [−.007,.007]	+0.367 [.319,.421]	+0.110 [.077,.145]
LSTM	−.001 [−.005,.000]	−.003 [−.010,.004]	+0.167 [.136,.198]	+0.025 [.003,.045]
State-space	+0.143 [.091,.194]	+0.121 [.082,.163]	+0.589 [.550,.628]	+0.194 [.153,.235]
Fixed-trust	+0.126 [.068,.185]	+0.245 [.185,.305]	+0.345 [.304,.388]	+0.171 [.131,.210]

At a fixed decision point, moving the cue from the shortest to longest registered offset reduces held-out pre-decision linear-readout performance for state-space and fixed-trust on all four tasks. The range is .121–.245 ordinary accuracy on Reacher and .171–.589 balanced accuracy on PushT; all eight paired intervals exclude zero. GRU and LSTM losses resolve on both PushT tasks but not on Reacher. This identifies a within-task cue-placement effect along the evidence-age axis under fixed base trajectories, not a universal monotonic forgetting law or a complete explanation of the cross-checkpoint reversal.

H.2 EXISTING-CHECKPOINT READ-TIME DIAGNOSTIC

As a descriptive precursor, we read the already-trained memory modules at earlier causal times without retraining. The state read always precedes the observation at that time and the last three raw latents are joined in the same way as the registered endpoint. Earlier reads show a pronounced transient trace for state-space and fixed-trust memory (Table below), while GRU/LSTM changes are much smaller. This branch does *not* identify delay: read time, physical state, and remaining episode length change together. The fixed-endpoint intervention above is the causal comparison used in the main paper.

Table 20: Existing-checkpoint paired accuracy gain over no state at earlier state reads. “Endpoint” is the registered pre-decision read (15 observations after the cue for PushT; 43–53 for Reacher).

Task	Module	Age 4	Age 8	Age 15	Endpoint
Reacher marker	State-space	+ .683	+ .447	+ .151	+ .014
	Fixed-trust	+ .676	+ .603	+ .313	+ .021
Reacher color	State-space	+ .678	+ .383	+ .121	+ .030
	Fixed-trust	+ .678	+ .547	+ .236	− .007
PushT token	State-space	+ .706	+ .363	+ .060	+ .060
	Fixed-trust	+ .761	+ .687	+ .396	+ .396
PushT binding	State-space	+ .303	+ .078	+ .013	+ .013
	Fixed-trust	+ .439	+ .219	+ .052	+ .052

I LONG-CONTEXT AND LEARNED-ROLLOUT CONTROLS

I.1 CONTEXT COVERAGE AND SEMANTIC EXPOSURE

Table 21: Five-seed long-context controls. Predictor entries are mean [95% seed interval]; validation MSE is the five-seed mean. Raw readouts are seed-independent.

Task	H	Raw-context accuracy	Predictor accuracy	Validation next-latent MSE
Marker	3	.229	.262 [.258,.266]	.0524
	16	.250	.255 [.252,.259]	.0132
	32	.263	.250 [.247,.253]	.0049
	56	.771	.253 [.250,.256]	.0065
Color	3	.171	.206 [.202,.210]	.0837
	16	.179	.188 [.180,.195]	.0588
	32	.196	.185 [.181,.189]	.0525
	56	.825	.199 [.190,.211]	.0663

At $H \in \{3, 16, 32\}$, no validation window contains a cue frame. At $H = 56$, every window contains cue evidence. The raw-access discontinuity therefore reflects legal coverage, while the flat predictor curve diagnoses learned exposure under next-latent training.

I.2 FIVE-SEED LEARNED-ROLLOUT CONTROL

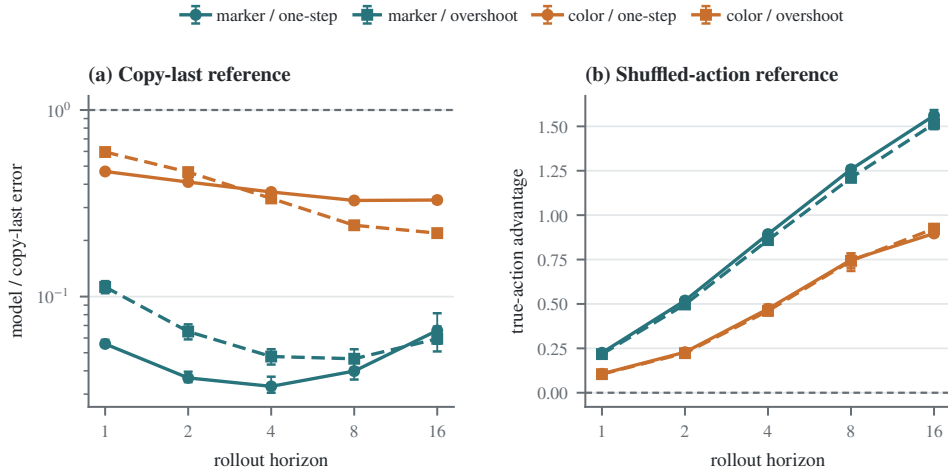


Figure 7: Learned rollout competence from the fixed local anchor. Ratios below one beat copy-last; positive action advantage beats shuffled actions. This control does not contain pre-anchor memory evidence.

Table 22: Five-seed rollout results at the registered horizon-eight gate. Ratios are model/copy-last normalized MSE; action advantage is shuffled-action MSE minus true-action MSE.

Task	Objective	Copy ratio [95% CI]	Action advantage [95% CI]	Passing seeds
Marker	one-step	.040 [.036,.047]	1.258 [1.240,1.276]	5/5
Marker	overshoot-8	.046 [.042,.052]	1.210 [1.186,1.228]	5/5
Color	one-step	.327 [.322,.334]	.748 [.701,.784]	5/5
Color	overshoot-8	.241 [.239,.243]	.742 [.686,.784]	5/5

All task/objective models are also evaluated at horizons 1, 2, 4, and 16. No rollout metric is included in the retention aggregate; complete values and seed arrays are stored in the authenticated extension summary.

J DIAGNOSTIC-ONLY DINO-WM PRECURSOR

We apply the same cue/shortcut/temporal logic to a second latent world-model family, the original official proprioceptive DINO-WM PushT checkpoint (Zhou et al., 2025). This is a protocol-portability check, not a matched carrier comparison. The released LeWM-bank `dinowm_noprop` URL was unavailable. A first preregistered attempt to run the original checkpoint on the LeWM bank stopped before task metrics because agent-position mean shift was .318 training standard deviations, above the .25 compatibility limit. The completed branch instead uses the checkpoint’s original OSF PushT distribution and keeps the failed compatibility record unchanged.

We reserve 1,200/480 disjoint episodes from the native training-distribution bank because only 16 official validation episodes are long enough for the fixed 20-frame, skip-five audit. The DINO encoder and world model remain frozen. Cue decoding uses a fixed spatial-pyramid feature interface; observed context, teacher-forced endpoint, actions, and proprioception must remain within .05 of chance. A three-frame cue anchors native open-loop imagination. At each of 1, 4, 8, and 15 imagined observations, the same predeclared readout class is fit separately on the training split and evaluated on held-out episodes. The rollout must beat copy-last at one step and in the 1–16 average and must beat episode-shuffled actions before the age-indexed diagnostics are evaluated.

Table 23: DINO-WM native-distribution prerequisite checks. Values are balanced accuracy except the final two rollout-health entries.

Check	Token (ceiling .300)	Binding (ceiling .217)
Cue decoding (minimum .750)	1.000	1.000
Observed-context shortcut	.185	.152
Teacher-endpoint shortcut	.298	.163
Action / proprioception shortcut	.219 / .265	.181 / .175
1–16 model/copy-last ratio (required < 1)		.200
1–16 true-action advantage (required > 0)		.790

Table 24: DINO-WM cue-anchored imagination. Readout intervals bootstrap held-out episodes. Transport is within-episode all-label counterfactual RMS at the requested age divided by cue-time RMS.

Task	Imagined observations since cue	Balanced accuracy [95% CI]	Transport ratio [95% CI]
Token	1	.992 [.983,.998]	.337 [.325,.349]
	4	.600 [.556,.644]	.420 [.383,.460]
	8	.265 [.225,.304]	.573 [.514,.634]
	15	.271 [.231,.310]	.721 [.651,.794]
Binding	1	.969 [.952,.983]	.517 [.506,.529]
	4	.365 [.323,.408]	.508 [.461,.560]
	8	.238 [.200,.277]	.821 [.722,.926]
	15	.183 [.148,.219]	1.058 [.931,1.200]

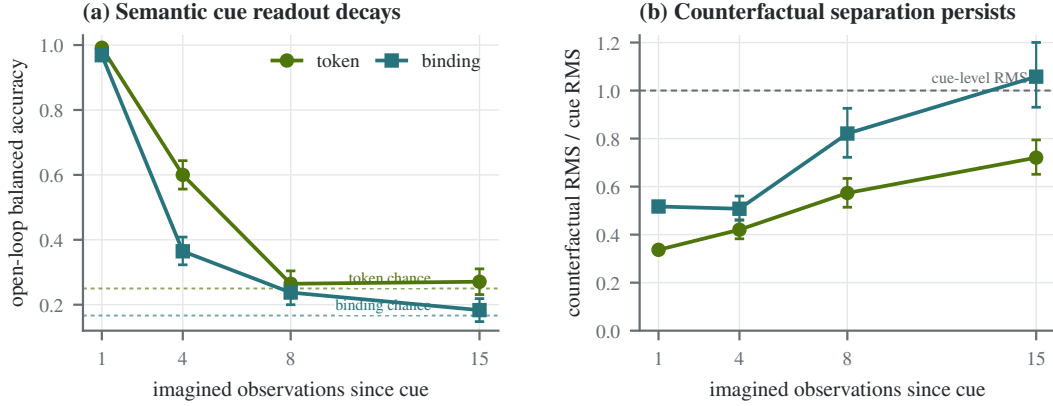


Figure 8: Age-indexed DINO-WM imagination diagnostics. Token recall is unresolved by age eight; binding remains above chance at age eight but is unresolved at age 15. Paired full-patch feature distance can grow as linear semantic decodability collapses, so raw counterfactual rollout divergence is not stable, usable memory.

This precursor extends the cue-availability, shortcut, rollout-health, and age-indexed imagination diagnostics beyond LeWM. It does not itself test persistent state, a carrier swap, or downstream use; the separately locked persistent-carrier extension appears in Appendix D. Full-patch distance may reflect nonspecific rollout divergence. Neither study is a numerical comparison with LeWM: the bank is native rather than matched, both precursor readout splits come from the checkpoint’s training distribution, and the token teacher-endpoint shortcut passes narrowly at .2979 against .3000.

K LEGACY FIXED-CONSUMER EXECUTION CONTROLS

Both tests ask whether an external consumer can turn retained evidence into a different executed outcome. They do not reuse validation labels for fitting, and they do not test a native LeWM planner. The Reacher consumer chooses a joint goal for a fixed PD controller; the PushT consumer chooses a categorical block-pose goal for one pinned native-physics controller.

K.1 REACHER DRIFTING-COLOR TEST

Table 25: Drifting-color fixed-consumer test. Means pool five representation checkpoints over the same 240 validation episodes. Regret is relative to the label oracle; lower is better.

Representation source	Goal decision accuracy	Executed success	Mean regret
No persistent carrier	.180	.175	.722
GRU	.190	.185	.712
State-space	.203	.196	.702
Fixed-trust	.178	.173	.724
Raw 56-latent context	.844	.782	.133
Cue-window representation	1.000	.917	.000

The paired state-space-minus-no-state success difference is .0208 [.0017,.0417], and the paired regret reduction is .0199 [.0010,.0410]. GRU and fixed-trust intervals cross zero. Label-shuffle and action/time controls pass for every representation source. Absolute state-space goal selection remains below four-way chance.

The marker task is ineligible for downstream inference because the cue-window physical oracle reaches .8875, below .90. A controller-only amendment evaluates four longer-horizon or more damped candidates on a class-balanced subset of the training bank. Every candidate reaches .8708 on marker and .9167 on color, below the locked per-task .925 development threshold. The amendment writes a stop record and never reads validation.

K.2 PUSH T TOKEN AND BINDING TESTS

The training split is divided without labels into 800 consumer-training and 400 development episodes; the original 480 validation episodes remain held out. A native block-position rule retains 409 held-out

episodes. One further episode is excluded before reading its label because simulator reset moves the block by 1.333 pixels, above the predeclared one-pixel fidelity limit. All reported results therefore use the same 408 episodes (85.0%).

For each task and carrier-training seed, one arm-blind logistic consumer is fit to a deck pooling all five representation sources. Arm identity is unavailable. Its feature is $[z_{16}; z_{17}; z_{18}; b_{19}]$, with observation 19 excluded. The selected class indexes a fixed radial block-pose goal; the same deterministic controller receives the cached simulator state only after selection. Thus only the selected goal differs across arms. Carrier features run on CUDA 1; execution uses a clean checkout of the native simulator at commit 0ef3856.

Table 26: PushT downstream prerequisite checks. Shortcut and shuffle ceilings are chance plus .05. Oracle values are executed-success rates on the development split.

Check	Token (4-way)	Binding (6-way)
Cue balanced accuracy / minimum class recall	1.000 / 1.000	.960 / .929
Largest late shortcut / ceiling	.264 / .300	.168 / .217
Mean of 20 label shuffles / ceiling	.257 / .300	.167 / .217
Physical oracle / minimum class success	.996 / .994	.994 / .988
Wrong-goal executed success	.000	.000

Table 27: Complete PushT fixed-consumer results. Selection and success are means over five carrier seeds and the same 408 episodes. Paired intervals use a 10,000-draw crossed seed-by-episode bootstrap. Pose regret is lower-is-better and measured in the controller’s native block-pose units.

Task	Representation	Selection	Success	Pose regret	Δ success vs. none [95% CI]
Token	No persistent state	.238	.238	140.93	—
	GRU	.246	.246	139.55	+ .008 [− .003, .020]
	LSTM	.244	.244	140.12	+ .006 [− .007, .019]
	State-space	.271	.271	134.55	+ .033 [.013, .057]
	Fixed-trust	.304	.304	127.92	+ .067 [.039, .096]
Binding	No persistent state	.158	.158	144.01	—
	GRU	.169	.166	142.66	+ .008 [.001, .017]
	LSTM	.158	.156	144.01	− .001 [− .011, .007]
	State-space	.175	.172	141.60	+ .014 [.002, .029]
	Fixed-trust	.175	.172	142.40	+ .014 [.001, .028]

The realized random-goal success is .243 for token and .181 for binding; the corresponding nominal expectations are .250 and .167. The held-out oracle is .995 for both tasks. On token recall, state-space and fixed-trust also reduce pose regret by 6.39 [1.88, 11.59] and 13.02 [6.82, 19.56]. Binding regret intervals cross zero, and both carrier success rates remain below the realized random baseline. We therefore credit token recall with an external-consumer execution effect and binding only with relative downstream sensitivity. Neither is standard PushT performance or evidence that LeWM’s native predictor or planner reads the memory state.

L LABEL-FREE REPAIR DIAGNOSTICS

The first repair reconstructs four evenly spaced frozen cue-time latents from the final carrier prior. Compared with equal continued training, neither GRU nor state-space repair has an interval-resolved executed success or regret gain. The second diagnostic predicts

$$\bar{z}_{\text{cue}} - \frac{1}{2} (z_{\text{pre}} + z_{\text{post}}) \quad (3)$$

from the final prior. The target is label-free, causal, standardized only on the formal training cache, and never optimized on validation.

Table 28: Cue-residual repair. Positive twin MSE reduction means the auxiliary head fits better than objective-off continued training. Support additionally requires held-out NMSE below the training-mean predictor.

Task	Carrier	Twin MSE reduction [95% CI]	NMSE / mean	Next-latent change
Marker	GRU	.970 [.928, 1.027]	1.012	+ .0025
Marker	State-space	1.164 [1.102, 1.230]	1.034	+ .0065
Color	GRU	1.347 [1.273, 1.425]	1.007	+ .0046
Color	State-space	2.420 [2.249, 2.609]	1.008	+ .0108

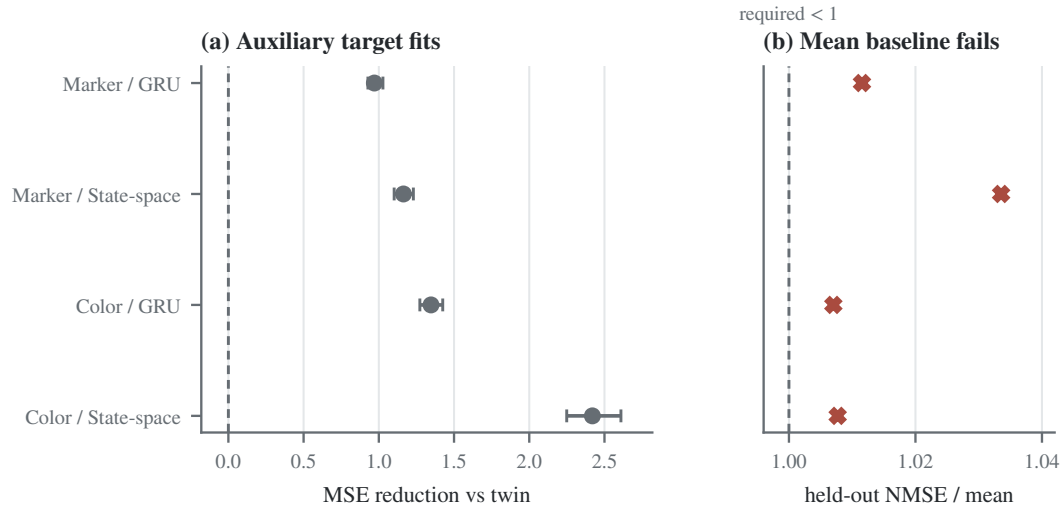


Figure 9: The auxiliary head reduces residual MSE versus its continued-training twin (a), but held-out NMSE remains above the training-mean baseline in every cell (b). The predeclared success criterion fails in all four cells.

M SHELL-GAME CAPACITY ATTEMPTS

The capacity study tracks one, two, or four colored items through visible cup swaps and queries final item slots. Paired counterfactuals, swap visibility, post-shuffle leakage, and late-context leakage are tested before carriers train. The initial cue and one versioned salience amendment fail frozen cue availability. A second development-only amendment with one predeclared large cue card passes at .900–.946 per item and is locked before formal data are read.

On the locked cue-card formal bank, cue availability is .954 and swap visibility is .971. The one-item stage passes all gates. The two- and four-item stages each yield .392 late-context accuracy on one item, above the locked chance-plus-.05 ceiling of .383. Because the grid is a capacity comparison, we do not selectively train only the favorable one-item stage after seeing the formal failures.

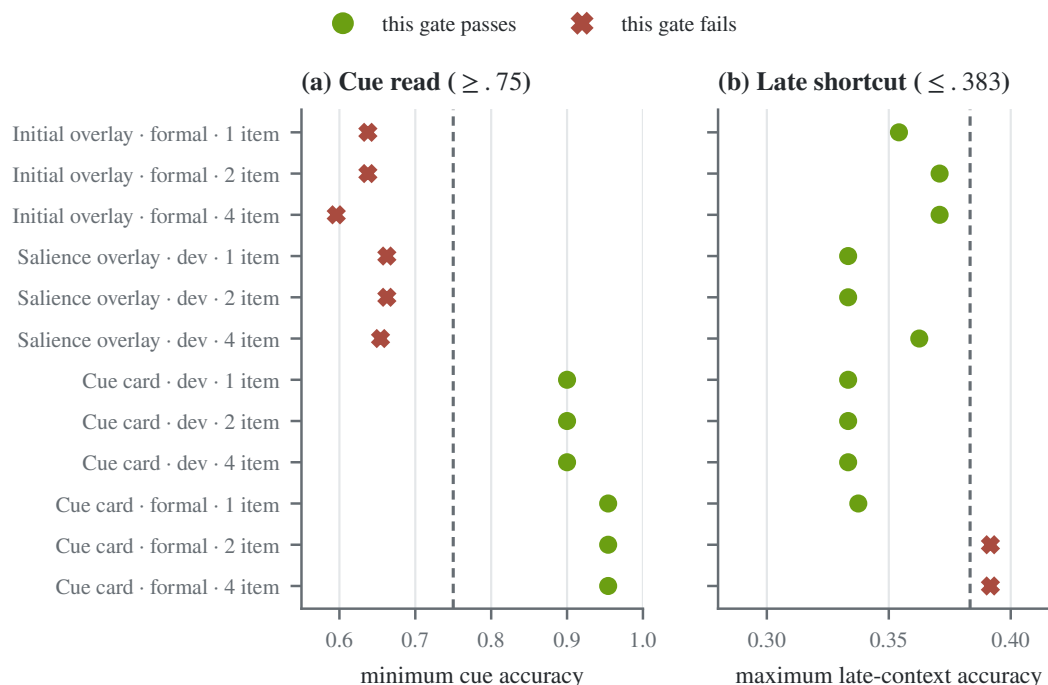


Figure 10: Complete shell-game gate history. The initial and salience overlays fail cue availability. The cue-card design passes in development, but its formal multi-item banks narrowly fail the no-shortcut gate; no memory-module comparison is produced.

N STATISTICAL CONTRACT AND REPRODUCIBILITY LEDGER

Inference units. The matched-color primary study jointly resamples carrier-seed indices across LeWM hosts, independently resamples episodes within host, and stratifies by the 16 color–location combinations. DINO-WM PushT resamples matched carrier seeds and class-stratified held-out episodes. PointMaze resamples matched carrier seeds and native validation episodes while keeping each four-label counterfactual set in one cluster. The legacy carrier intervals resample matched optimizer/model seeds; fresh Reacher banks are averaged with tasks inside seed and are not treated as independent runs. Fixed-consumer intervals use a crossed paired bootstrap over checkpoint seeds and validation episodes. Residual-repair intervals pair condition twins by checkpoint seed and episode. No-carrier copies are never counted as independent seeds.

Table 29: Legacy and precursor artifact ledger. Hashes are SHA-256 prefixes shown for readability; full digests are stored in the corresponding locks and summaries. The matched and persistent DINO-WM grids use their separately generated, receipt-bound result tables and manifests.

Study	Formal cells	Spec/summary hash prefix	Integrity checks
Reacher carrier robustness	30 train + 100 eval	10637be6 / 7ca5b62	parent states, fresh banks, endpoints
Official PushT carrier swap	50 train	b7ac425a / 46400257	dataset, checkpoint, caches, frozen host
Reacher consumer + first repair	40 train + 10 eval	3e2f6f27	controls, oracle, crossed pairs
Cue-residual repair	40 train	c7a60d16	condition twins, causal target, no labels
Context/rollout extension	24 new train	b97dc9eb / 96dd17e9	36 parent + 24 new cells
Fixed-endpoint cue placement	80 train + 20 no-state eval	c8f6ac92 / 01ec607d	paired ages, 300 hashes, receipt 07c6581c
PushT external-consumer execution	50 representation cells	3cf007e2 / 38847fe9	gates, reset fidelity, 128 hashes
DINO-WM native portability	2 tasks × 4 ages	142240cb / 6c34f5b	native bank, admissions, rollout health
Shell-game cue card	admission only	8dc779da	versioned development and formal gates

All official weights are loaded strictly, and their full parameter-state digest is checked before and after every carrier job. Derived summaries validate every expected path and reject missing, unexpected, staged, or hash-mismatched artifacts. The official PushT HDF5 contains 18,685 episodes and 2,336,736 rows; its extracted SHA-256 begins b6ebd9ac. Formal GPU jobs use only CUDA

devices 0–2. Figure PDFs use embedded TrueType fonts and are generated from authenticated summaries and source/check artifacts.

O LEGACY CLAIM BOUNDARY

For the heterogeneous-cue, context, rollout, and repair studies above, the evidence establishes: (i) frozen cue encoding for four categorical tasks; (ii) causal final-read carrier comparisons under two LeWM checkpoints; (iii) robustness to additional Reacher seeds and fresh banks; (iv) a fixed-endpoint cue-timing intervention; (v) a raw-context/predictor-output dissociation; (vi) external-consumer execution effects on Reacher color and PushT token, plus relative PushT binding sensitivity; (vii) partial diagnostic portability to native DINO-WM imagination; and (viii) two unsuccessful label-free repairs. The controlled matched and persistent DINO-WM studies have separate task-level claims and receipt-bound tables.

It does not establish native consumption of state by a released LeWM planner, end-to-end memory-conditioned control, policy optimality, simultaneous multi-item capacity, natural-scene memory, universal carrier rankings, or carrier generalization across world-model families.